

Polymorphism

Concepts of Programming Languages
Lecture 4

Practice Problem

```
let k = fun x -> fun y -> x in  
let x = 3 + k (k 2 3) 4 in  
k x (k x)
```

Would UTop throw a type error on the above expressions? If not, what does it evaluate to?

Answer

let k = fun x -> (fun y -> x) in
let x = 3 + k k 2 3 4 in
k x (k x)

$\rightarrow$
let x = 3 + (fun x -> (fun y -> x)) (fun x -> fun y -> x)

$\rightarrow$
let x = 3 + [fun x -> fun y -> x / x] (fun y -> x) 2 3 4 in

$\equiv$
let x = 3 + (fun y -> (fun x -> fun y -> x)) 2 3 4 in

$\rightarrow$
let x = 3 + [2 / y] (fun x -> fun y -> x) 3 4 in

$\equiv$
let x = 3 + (fun x -> fun y -> x) 3 4 in

Answer

```
let k = fun x -> fun y -> x in  
let x = 3 + k k 2 3 4 in  
k x (k x)
```

$\text{let } x = 3 + (\text{fun } x \rightarrow \text{fun } y \rightarrow x) \ 3 \ 4 \ \text{in}$

$\rightarrow$
 $\text{let } x = 3 + [3/x](\text{fun } y \rightarrow x) \ 4 \ \text{in}$

$\rightarrow$
 $\text{let } x = 3 + (\text{fun } y \rightarrow 3) \ 4 \ \text{in}$

$\rightarrow$
 $\text{let } x = 3 + 3 \ \text{in} \ k \ x \ (k \ x)$

$\rightarrow$
 $k \ 6 \ (k \ 6)$

Answer

```
let k = fun x -> fun y -> x in
let x = 3 + k k 2 3 4 in
k x (k x)
```

$k \ 6 \ (k \ 6)$

$\rightarrow (fun \ x \rightarrow fun \ y \rightarrow x) \ 6 \ ((fun \ x \rightarrow fun \ y \rightarrow x) \ 6)$

$\rightarrow (fun \ x \rightarrow fun \ y \rightarrow x) \ 6 \ (fun \ y \rightarrow 6)$

$\rightarrow [6/x] (fun \ y \rightarrow x) \ (fun \ y \rightarrow 6)$

$\rightarrow (fun \ y \rightarrow 6) \ (fun \ y \rightarrow 6)$

$\rightarrow [fun \ y \rightarrow 6/y] \ 6 \ = \underline{\underline{6}}$

Outline

- » Discuss **polymorphism** in general
- » Demo **examples** of polymorphic functions

Polymorphism

Recall: Polymorphism and Lists

```
let rec length l =  
  match l with  
  | [] -> 0  
  | x :: xs -> 1 + length xs
```

What is the type of the length function?

Does this function depend on the values in the list?

Recall: The List Type

[1;2;3]

int list

["1";"2";"3"]

string list

[[1;1];[2;2];[3;3]]

int list list

The list type is an example of a **parametrized** type

A function on lists is *polymorphic* (with respect to the list parameter) if it can be apply to a list parametrized by *any* type

For this, we need *type parameters* to stand for *any* type:

'a, 'b, 'c, ...

Intuition: Int List Reversal

Implement the function

reverse : int list -> int list

*such that **reverse** **l** is the same as **l** but in reverse order*

Implementation

```
let rec rev_int l =  
  match l with  
  | [] -> []  
  | x :: tl -> (rev_int tl) @ [x]  
  
let _ = assert (rev_int [1;2;3] = [3;2;1])
```

Take the first element of l

Reverse the tl

Append [x] to the end of reversed tl

Intuition: String List Reversal

Implement the function

reverse : string list -> string list

*such that **reverse** l is the same as l but in reverse order*

Implementation

```
let rec rev_string l =  
  match l with  
  | [] -> []  
  | x :: tl -> (rev_string tl) @ [x]  
  
let _ = assert (rev_string ["1";"2";"3"] = ["3";"2";"1"])
```

Take the first element of `l`

Reverse the `tl`

Append `[x]` to the end of reversed `tl`

What's the Similarity?

```
let rec rev_int l =  
  match l with  
  | [] -> []  
  | x :: tl -> (rev_int tl) @ [x]
```

```
let rec rev_string l =  
  match l with  
  | [] -> []  
  | x :: tl -> (rev_string tl) @ [x]
```

Exact same implementation!

Copy/pasting code is *time consuming* and *error prone*

Polymorphism allows for better code reuse. The *same* function can be applied in multiple contexts

Not all functions are polymorphic

```
let rec sum l =  
  match l with  
  | [] -> 0  
  | x :: xs -> x + sum xs
```

Can this function be applied to a list parametrized by any type?

Answer: No, it can only be applied to **int lists**

OCaml's type inference is good at "guessing" when functions are polymorphic

Basic Example

```
let id = fun x -> x
let a = id 0
let b = id (0 = 0)
let c = id id
```

We want to be able to define functions that can be used in multiple contexts *and* that we can type check

Important: We can evaluate this if we *don't* type check

*But if we type-check, what should be the type of **id**?*

Answer: Parametric Polymorphism

```
let id = fun x -> x
let a = id 0
let b = id (0 = 0)
let c = id id
```

Parametric polymorphism allows for functions which are agnostic to the types of its inputs (this is what OCaml does)

For example, we can write a single identity function and use it with multiple types

Type Parameters

```
let id : 'a -> 'a = fun x -> x
```

The "parametric" part is the fact that types have *parameters*

Type parameters are instantiated at particular types according to the context

Polymorphism and Type Inference

```
fun f -> fun x -> f (x + 1)
```

In OCaml, the type of an expression depends on how its subexpressions are *used*

OCaml can figure out what is the "most" polymorphic (general) type we can give to an expression

This is called **type inference**

There are many subtleties
to this...

One Subtlety: Dynamic Dispatch

```
let to_string (x : 'a) : string = ...  
(* This is not possible in OCaml *)
```

Parametric polymorphism cannot be used for *dispatch*

We can't write a polymorphic function that "checks the type" to see what to do

Practice Problem

```
let rec f x = f (x + 1) in f
```

What is the type of the above OCaml expressions?

Answer

```
let rec f x = f (x + 1) in f
```

Practice Problem

```
let rec f x = f (f (x + 1)) in f
```

What is the type of the above OCaml expressions?

Answer

```
let rec f x = f (f (x + 1)) in f
```

Aside: Parametricity

$(\text{fun } x \rightarrow \text{fun } y \rightarrow x)$
 $(\text{fun } x \rightarrow x)$

*Can you write function of type **'a** -> 'a?*

Can you write another?

Parametricity refers to the (very cool) observation that polymorphic types often restrict the kinds of functions you can write in a *mathematically rigorous sense*

Challenge: Parametricity in Lists

Can you write function of type:

'a list -> 'a list?

How many such functions can you write?

What can these functions do?

Summary

Polymorphism allows functions to be agnostic to (parts of) the types of its parameters

Parametric polymorphism does not allow for dynamic dispatch, it must given the *same* implementation at any particular type